

RAHEEB HASSAN

PhD Student in Computer Science, UC Irvine

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RESEARCH INTERESTS

Machine Learning, Neurosymbolic AI, Reinforcement Learning and World Models

EDUCATION

- 2025 – **University of California, Irvine** *Ph.D. in Computer Science*
Donald Bren School of Information and Computer Sciences
Advisor: Dr. Mohsen Imani
Lab: BIASLab
- 2018 – 2023 **University of Dhaka** *BSc. (Hons) in Computer Science and Engineering*
Department of Computer Science and Engineering
CGPA: 3.75 (2nd highest among 54 students)
Thesis Topic: Safe Multi-Agent Reinforcement Learning
Supervisor: Dr. Md. Mosaddek Khan

RESEARCH EXPERIENCE

- 2025 – **Doctoral Researcher** *BIASLab, University of California, Irvine*
 - Investigating **multimodal anomaly detection** and recognition
 - Developing **graph neural network** methods for anomaly detection from image and audio data
 - Researching **hyperdimensional computing** and **neurosymbolic** methods
 - Working on **reinforcement learning** and **world models**
- 2021 – 2023 **Research Assistant** *Cognitive Agents and Interaction Lab (CAIL), University of Dhaka*
 - Studied and benchmarked several **reinforcement learning** algorithms
 - Researched **safety** in reinforcement learning and **multi-agent systems**
 - Developed a novel decentralized safe multi-agent reinforcement learning algorithm (DePAint)
 - Worked with applying **graph neural networks** and reinforcement learning for solving combinatorial optimization problems

TEACHING EXPERIENCE

- 2026 – **Graduate Teaching Assistant** *University of California, Irvine*
 - ICS 26:** Data Structures Implementation and Analysis
 - Office hours, review sessions, grading and mentoring

PROFESSIONAL EXPERIENCE

- 2023 – 2025 **Machine Learning Engineer** *Therap Services LLC*
Associate ML Engineer (2023 – 2024), ML Engineer (2024 – 2025)
 - Worked with **computer vision**, **face recognition**, **multi-object tracking**, **activity recognition**
 - Designed a system integrating real-time monitoring data from cameras and sensors to generate long-term statistical models of individual behavior
 - Optimized ML pipelines to work on resource constrained environments using **NVIDIA DeepStream**
 - Worked on scaling distributed ML video processing systems in the cloud (**AWS**, **Oracle Cloud**)
- 2020 – 2021 **Full Stack Developer** *CranTech LLC, New York, USA*
Remote, part-time
 - Worked on several web development projects like **Shoshikkha**, **Amplify**, etc.
 - Worked with **ReactJS** (frontend), **ExpressJS** (backend) and **React Native** (Mobile)

SKILLS

- Languages:** Python, C++, C, Java, Javascript, Typescript, C# (Unity3D), Dart (Flutter).
- Machine Learning:** PyTorch, TorchVision, PyTorch Geometric, NVIDIA DeepStream, Scikit-Learn, Optuna, Pandas, NumPy, Matplotlib, Seaborn.
- Miscellaneous:** Git, Docker, Linux (Debian, Arch), AWS Cloud, Oracle Cloud, ReactJS, NodeJS, FastAPI, Flask.

PUBLICATIONS

Accepted 2026	Qubit-Efficient Quantum Search for Hyperdimensional Decomposition via Logarithmic Encoding <i>IEEE/ACM International Conference on Computer-Aided Design (ICCAD) 2026</i> Authors: Sanggeon Yun, Hyunwoo Oh, Ryoza Masukawa, Raheeb Hassan , Mohsen Imani Preprint: arxiv.org/abs/2607.11936
Published 2026	Contextual Fusion Strategies for Multimodal GNN-Based Reasoning: Performance and Computational Trade-Offs <i>IEEE Access</i> Authors: Sanggeon Yun, Ryoza Masukawa, Raheeb Hassan , Minhyoung Na, Mohsen Imani DOI: 10.1109/ACCESS.2026.3653660
Published 2024	Reinforcement Learning Based Formulations with Hamiltonian-Inspired Loss Functions for Combinatorial Optimization over Graphs <i>IEEE Access</i> Authors: Redwan Ahmed Rizvee, Raheeb Hassan , Md. Mosaddek Khan DOI: 10.1109/ACCESS.2024.3497955 / Preprint: arxiv.org/abs/2311.16277
Published 2024	DePAint: A Decentralized Safe Multi-Agent Reinforcement Learning Algorithm considering Peak and Average Constraints <i>Springer Applied Intelligence</i> Authors: Raheeb Hassan , K.M. Shadman Wadith, Md. Mamun or Rashid, Md. Mosaddek Khan DOI: 10.1007/s10489-024-05433-x / Preprint: arxiv.org/abs/2310.14348

Preprints

2026	CyberWorld: World Models for Sample-Efficient Autonomous Cyber Defense <i>arXiv</i> Authors: Ryoza Masukawa, Sanggeon Yun, Raheeb Hassan , Hyunwoo Oh, Sungheon Jeong, Mohsen Imani Preprint: arxiv.org/abs/2609.31893
2026	Vector Symbolic Policy Gradient <i>arXiv</i> Authors: Ryoza Masukawa, Sanggeon Yun, Sungheon Jeong, Hyunwoo Oh, Raheeb Hassan , Pietro Mercati, Nathaniel D. Bastian, Mahdi Imani, Mohsen Imani Preprint: arxiv.org/abs/2608.18404 / Code: github.com/BiasLabProjects/VSPG
2026	n-Musketeers: Reinforcement Learning Shapes Collaboration Among Language Models <i>arXiv</i> Authors: Ryoza Masukawa, Sanggeon Yun, Hyunwoo Oh, Sungheon Jeong, Raheeb Hassan , Hanning Chen, Wen-Jun Huang, Mahdi Imani, Pietro Mercati, Nathaniel D. Bastian, Mohsen Imani Preprint: arxiv.org/abs/2602.09173
2026	HopFormer: Sparse Graph Transformers with Explicit Receptive Field Control <i>arXiv</i> Authors: Sanggeon Yun, Raheeb Hassan , Ryoza Masukawa, Sungheon Jeong, Mohsen Imani Preprint: arxiv.org/abs/2602.02268
2025	MissionHD: Hyperdimensional Refinement of Distribution-Deficient Reasoning Graphs for Video Anomaly Detection <i>arXiv</i> Authors: Sanggeon Yun, Raheeb Hassan , Ryoza Masukawa, Nathaniel D. Bastian, Mohsen Imani Preprint: arxiv.org/abs/2508.14746

AWARDS AND ACHIEVEMENTS

Awards

Dean's Award
Faculty of Engineering & Technology, University of Dhaka

Best Undergraduate Final Project (Thesis) Award
Department of Computer Science and Engineering, University of Dhaka

Competitions

Hackathons

- Code Samurai: Inter-University Hackathon 2022 (**Champion**)

Programming Competitions

- International Collegiate Programming Contest Asia Dhaka Regional 2021 (**16th place**)
- MIST Inter University Programming Contest 2019 (**20th place**)
- SUB Inter University Programming Contest 2019 (**28th place**)

Cybersecurity Competitions

- SUST Technovent 2023 Capture the Flag (**11th place**)

Miscellaneous

Online Judge
Codeforces (expert) (**top 20%**) , Codechef (blue), LightOJ (90+ problems)